10.1 Homework

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Question 1

0/1 pt

Using the Law of Sines to solve the triangle if $\angle A = 44^\circ$, $\angle C = 74^\circ$, $b = 15$:

$\angle B$ is 47 degrees;

$a =$ 2.58 ;

$c =$ 21.77 ;

You may round to two decimal places.
Assume $\angle A$ is opposite side a, $\angle B$ is opposite side b, and $\angle C$ is opposite side c.

Question Help: Video

Question 2

1/1 pt 3-4

Using the Law of Sines to solve the all possible triangles if $\angle A = 105^\circ$, $a = 28$, $b = 11$.
If no answer exists, enter DNE for all answers.

$\angle B$ is 22.27 degrees;

$\angle C$ is 52.73 degrees;

$c =$ 23.09 ;

Round to two decimal places as needed.
Assume $\angle A$ is opposite side a, $\angle B$ is opposite side b, and $\angle C$ is opposite side c.

Question Help: Worked Example 1

Question 3

1/1 pt 3

Given a triangle with angle A = 96 degrees and sides a = 34 and b = 15. Find side c.

c = 28.98

Round to two decimal places as needed.

Question Help: Worked Example 1

Question 4

1/1 pt 4

Using the Law of Sines to solve the triangle if $\angle A = 35^\circ$, $\angle C = 74^\circ$, $b = 25$

$\angle B$ is 71 degrees

$a =$ 15.16

$c =$ 25.41

Assume $\angle A$ is opposite side a, $\angle B$ is opposite side b, and $\angle C$ is opposite side c.

Question 5

0.67/1 pt 0-4

Using the Law of Sines to solve for all possible triangles if $\angle B = 50^\circ$, $a = 103$, $b = 50$.
If no answer exists, enter DNE for all answers.

$\angle A$ is 65.27 degrees

$\angle C$ is DNE degrees

$c =$ DNE

Assume $\angle A$ is opposite side a, $\angle B$ is opposite side b, and $\angle C$ is opposite side c.

Question Help: Video

Question 6

0/1 pt

If $c = 201$, $\angle A = 75^\circ$ and $\angle B = 45^\circ$,

$b =$ 1.89 ;

Assume $\angle A$ is opposite side a, $\angle B$ is opposite side b, and $\angle C$ is opposite side c.

Question Help: Video

Question 7

1/1 pt 2-4

Use the Law of Sines to find a triangle **with all acute angles** if $b = 124$, $c = 169$, $\angle B = 40^\circ$.

$a =$ 189.25518 ;

$\angle A =$ 78.83 degrees;

$\angle C =$ 61.17 degrees;

Assume $\angle A$ is opposite side a, $\angle B$ is opposite side b, and $\angle C$ is opposite side c.

Question 8

1/1 pt 3-4

Use the Law of Sines to find a triangle with **one obtuse angle** if $\angle A = 49^\circ$, $a = 26$, $b = 29$.
If no answer exists, enter DNE for all answers.

$\angle B$ is 122.67 degrees;

$\angle C$ is 8.33 degrees;

$c =$ 4.99 ;

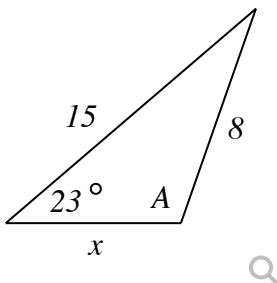
Assume $\angle A$ is opposite side a, $\angle B$ is opposite side b, and $\angle C$ is opposite side c.

Question 9

0/1 pt

Given the triangle below, find the angle *A* and length of side *x*

Note: picture is NOT drawn to scale, but you can assume an angle that appears acute is acute and angle that appears obtuse is obtuse.



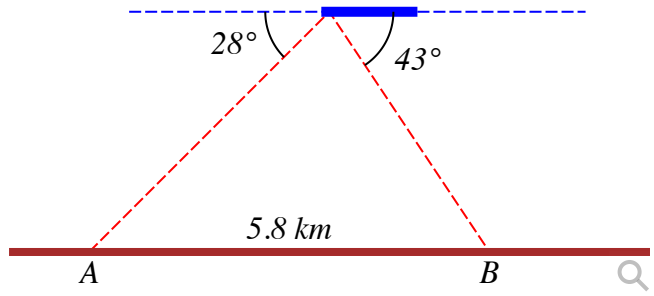
A = 47.1 degrees

x = 16.17

Question 10

1/1 pt 4

A pilot is flying over a straight highway. He determines the angles of depression to two mileposts, 5.8 km apart, to be 28° and 43°, as shown in the figure.



NOTE: The picture is NOT drawn to scale.

Find the distance of the plane from point *A*.

distance from *A* = 4.18 km

Find the elevation of the plane.

height = 1.96 km

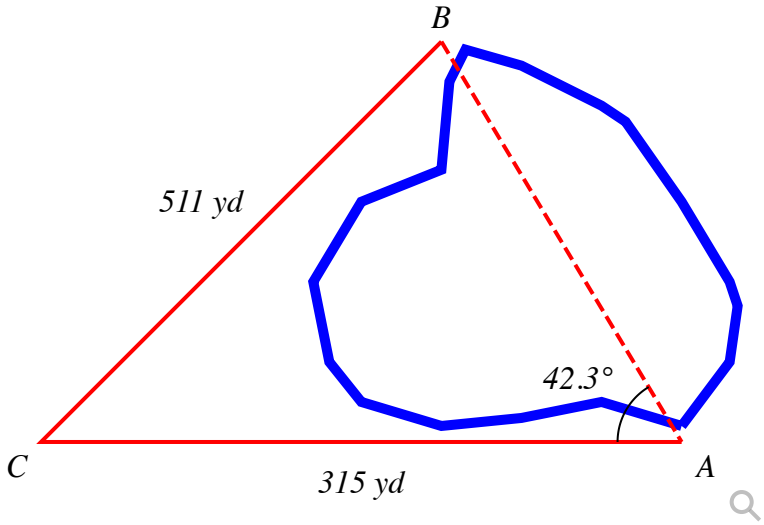
Enter your answer as a number; your answer should be accurate to 2 decimal places.

Question Help: Worked Example 1

Question 11

1/1 pt 1

Points *A* and *B* are separated by a lake. To find the distance between them, a surveyor locates a point *C* on land such than $\angle CAB = 42.3^\circ$. Find the distance across the lake from *A* to *B*.



NOTE: The triangle is NOT drawn to scale.

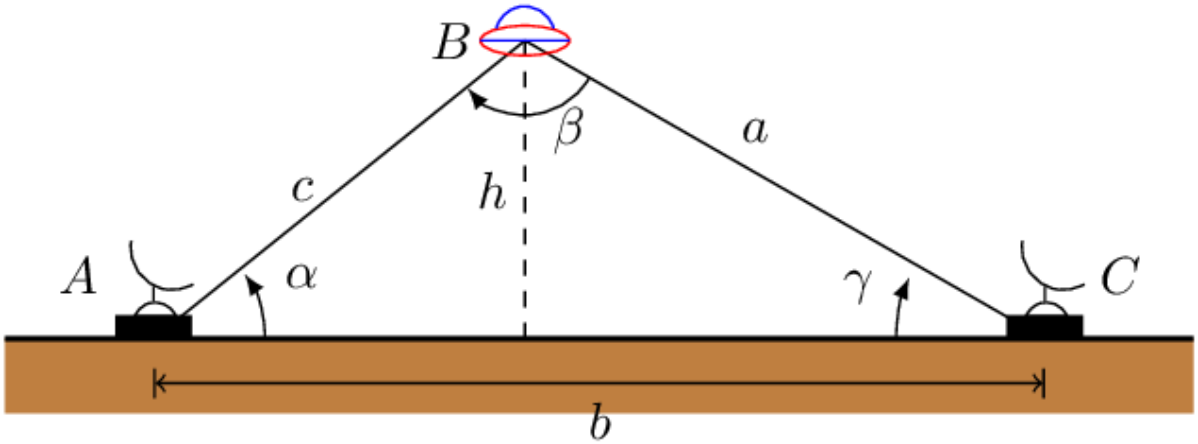
distance = 697.96 yd

Enter your answer as a number; your answer should be accurate to 2 decimal places.

Question Help: Worked Example 1

Question 12

0.5/1 pt 0-4



NOTE: The picture is NOT drawn to scale.

Ancient alien theorists manning two radar stations located b = 59 mi apart both detect a UFO located between them. The angle of elevation measured by the first station (A) is $\alpha = 87.8^\circ$ and the angle of elevation measured by the second station (C) is $\gamma = 84.7^\circ$.

How far is the UFO from station A?

distance from *A* = 450.08 mi

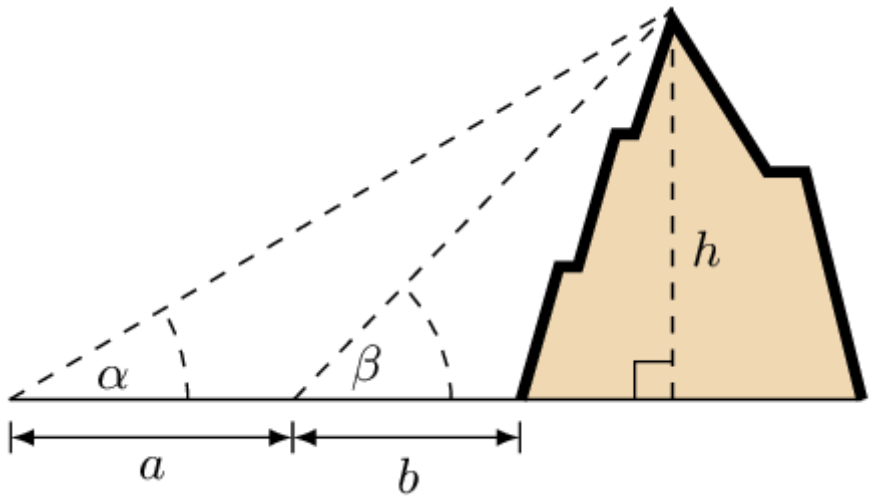
How high is the UFO above the ground?

h = -2.87 mi

Enter your answer as a number; your answer should be accurate to 2 decimal places.

Question Help: Video

Question 13 1/1 pt



To estimate the height of a mountain, two students find the angle of elevation from a point (at ground level) $b = 600$ meters from the base of the mountain to the top of the mountain is $\beta = 45^\circ$. The students then walk $a = 2700$ meters straight back and measure the angle of elevation to now be $\alpha = 39^\circ$. If we assume that the ground is level, use this information to estimate the height of the mountain.

The height of the mountain is 11494.39 meters.

Question Help: Video